

Problem Statement #34

The Smart Mechanical Fault Diagnosis Agent

The Challenge

Mechanical systems such as engines, pumps, and gearboxes often experience faults that are difficult to diagnose quickly. Engineers and technicians typically rely on manual inspections, experience, and trial-and-error troubleshooting methods. This process can lead to extended downtime, higher maintenance costs, and inefficient repair processes.

The Objective

Build an AI-powered Mechanical Fault Diagnosis Agent that analyzes machine symptoms, sensor data, and historical maintenance logs to identify possible mechanical faults.

Core Requirements

- **1. Fault Data Retrieval (RAG)** – Collect historical fault records and maintenance logs.
- **2. Symptom Analysis** – Analyze vibration patterns, noise signals, and temperature anomalies.
- **3. Root Cause Identification** – Identify the most probable cause of failure.
- **Repair Guidance** – Provide recommended repair procedures.

4. Multi-Agent System

- **Symptom Detection Agent** – Collects machine symptoms and sensor signals.
- **Fault Analysis Agent** – Matches symptoms with known fault patterns.
- **Root Cause Agent** – Determines the probable mechanical issue.
- **Repair Guidance Agent** – Suggests troubleshooting and repair procedures.

5. Visualization Dashboard

Display fault alerts, root cause explanations, and repair recommendations.

6. Tech Stack

IBM Langflow / IBM Orchestrate + IBM Models

Project: VitalMech — AI-Powered Mechanical Health & Fault Diagnosis